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1. Introduction

The research strategy provides the overall plan for the research. The strategy will guide the creation of objectives, data collection and data analysis.

“A Research Strategy is a step-by-step plan of action that gives direction to your thoughts and efforts, enabling you to conduct research systematically and on schedule to produce quality results and detailed reporting” (Dinnen, 2014).

The choice of strategy is linked to the aim of the research. For example, should the research aim to develop a new or improved AI algorithm, then design research would be a more suitable choice than Survey. If the research aims to understand the practices of a unique software engineering team then a Case Study would be more appropriate than an experiment. A survey is useful for measuring the attitude of an entire population.

There are numerous research strategies. For this module, we will focus on four popular types in IT research.

2. Survey

A survey is the process of designing and distributed questionnaires to participants. The data collected from surveys is then statistically analysed to identify patterns and trends and eventually arrive at a conclusion.

A survey is intended to collect data from a large *sample* of participants. The findings of the study from the sample is then used to generalize across a larger *population*. In other words, this information or opinion collected from the sample is more often a *generalization* of what a large population thinks. Patterns are identified in the collected data which contribute towards the results of the research. The researcher uses information from a sample to make some *inference* about the wider population.

Researchers do not attempt to control conditions or manipulate variables under investigation.

A survey does not have to rely solely on an online questionnaire. Data could emanate from project documentation such as, product backlogs; observation of task completion; journal publications or interviews. The literature review is in essence a type of survey.

Two main types of survey

The two main types of survey are online questionnaire or interview. An online questionnaire is easily developed using online tools such as QuestionPro and SurveyMonkey. Some tools have built in statistical analysis that provides results instantly. This can negate the need for a statistician. This is a far superior method of collecting data when compared to printed and posted or hand-delivered questionnaires primarily due to the shorter time required to complete the survey and analyse results. There is less room for mistakes and lost questionnaires. Professional looking surveys can be setup very quickly using online templates.

Interviews are usually conducted using a secure tool such as MS Teams. This is a preferred means of interaction when compared to face-to-face encounters. The IT industry has been using online collaborative tools for many years for communication and task completion. Hence, they are very familiar and sometimes more comfortable with this mode of interaction.

How are surveys conducted?

There are two main ways of deploying surveys, longitudinal and cross-sectional. A cross-sectional survey is used to collect data relevant to that period in time. It allows for the researcher to determine the state of affairs in that moment in time. These surveys are short and focus on one aspect of research.

Longitudinal survey involves the collection of data over a long period of time (5 years to 5 decades). There are three main types of longitudinal survey trend, panel and cohort.

Trend surveys are used to show how participants' opinions change over time. For example, what was the first year IT student's opinion of online LMS in their first year of study compared to now in their fourth year.

A researcher will use a *panel survey* when data is collected from the same sample over time. For example, the same sample of students from first year to fourth year. This allows for a more accurate telling of the change over time as the same participants are used.

A *cohort survey* is when the participants are selected for the sample using certain selection criteria. An example might be selecting students that entered the first year from school quintiles 1 to 3.

Basic steps in conducting a survey

Below is a generic process for conducting a survey.

1. Determine who will participate in the survey. This will require choice of target population and sampling technique.
2. Decide the type of survey. Online questionnaire or virtual interview.
3. Design the survey questions and layout. Using online tools and templates, the questionnaire can be designed using a selection of closed-ended questions such as, binary answers, likert scale, single answer option and multi answer option. Interview questions need to be planned to control the *flow* of responses and include prompts to focus on certain aspects.

4. Distribute the survey. The researcher must determine before hand how many *responses* are required in order for the data analysis to be valid. Determine how the survey will be advertised and volunteers contacted.
5. Analyze the responses. Data analysis can be done via several methods. These methods can be qualitative or quantitative.
6. Write up the results. Depending on the research work of either journal article, technical paper or dissertation, the research needs to be described.

Some advantages of using surveys

Online surveys are quick to develop and administer. The analysis of collected data is faster due to built in statistical tools. A wide reach is achievable. Participation can be controlled according to the above.

Some disadvantages of using surveys

The type of responses can be limited by the design or template used in the questionnaire. The design of the questionnaire has to be correct otherwise the incorrect data will be collected. Certain data can be missed out if there are no questions aimed at collecting that data.

Online Survey Tools

Surveys are conducted using online tools. There are several advantages to this which includes:

- Templates: a library of pre-designed questionnaire templates which can be matched to the type of survey
- Survey logic: navigates the participant through the survey avoiding unnecessary questions
- Flexible design: lets the researcher change colours, fonts, images and so on
- Data analysis: performs analysis of the collected responses and presents the results in a visual format
- Multi-language: provides support for switching the language of the questions instead of having to retype them

The following are among the more popular tools for creating online surveys

- SoGoSurvey
- Surveymonkey
- Google forms
- Survey planet

3. Design research

Design research is concerned with the creation of ICT based products, systems and solutions that respond to the needs of society and industry that are experiencing a persistent problem. It aims to improve on current problematic situations or provide alternatives for social interaction, education, finance and other areas of the economy.

Unlike the other strategies, design research does not rely completely on the collection and analysis of data. Instead, the focus is in *practical research*. In *pure research*, the emphasis lies in creating new knowledge and advancing the field. *Applied research* is done to achieve a real-world goal such as, improved quality of LMS for Higher Education.

Design science research aims to create new artefacts or theory via two main activities.

- The creation of new knowledge through the design of novel or innovative artefacts
- Analysis of the performance of the new artefact through use

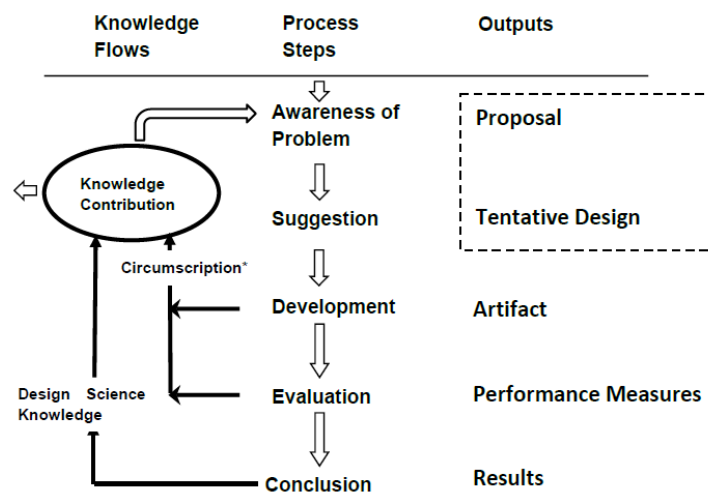
(Vaishnavi, 2019)

Typical examples of artefacts include, user interface heuristics, algorithms for search engines and cybersecurity, system development methodologies (Scrum).

Generic process flow for design science research

The figure below shows the stages of design science research.

Figure 1: Stages of design science research from Vaishnavi (2019).



Awareness of problem. Involves reading of relevant literature and interaction with stakeholder who have a vested interest in finding a solution to a practical problem. The end result is a proposal.

Suggestion. The envisioning of plans for improvement which can include new elements to be added or exiting ones to be modified. Prototypes are a popular technique. This stage lies in between the proposal and the design of the artefact. The end result is a design of the new artefact.

Development: iterative in nature. The artefact is created. This artefact can be a model, framework, systems process or new design theory. There are several methodologies that can be utilized depending on the type of artefact being created. The focus is on the novelty of the design of the artefact and not so much on the creation of the artefact.

Evaluation. Once the artefact is created, it needs to be evaluated for performance. This may involve analytical tools or human perception for fit-for-use. Techniques such as focus groups (e.g. new UX design) or performance benchmarking (e.g. a new ensemble machine learning algorithm) are used.

Conclusion: completion of the research. This may include writeup of the research, publication and packaging of the new design for patent or marketing.

It is important to note the flow back to previous stages for improved understanding, design or implementation of the artefact. This is not to be confused with the systems development life cycle used for software development. Instead, design science research is focused on the *design* aspect for an improved artefact.

4. Experiment

An experiment is used to determine the relationship between two aspects that are of interest in a research study. For example, a researcher might be interested in how a chatbot might assist the first year university student in various administrative processes. In this case the chatbot and the ability to correctly complete administrative processes are the *variables* of the research.

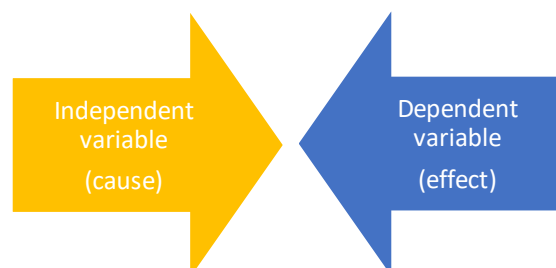
Experimental research seeks to determine a relationship between 2 variables, the dependent variable and the independent variable

“Experimental research is a study that strictly adheres to a scientific research design. It includes a hypothesis, a variable that can be manipulated by the researcher, and variables that can be measured, calculated and compared. Most importantly, experimental research is completed in a controlled environment. The researcher collects data and results will either support or reject the hypothesis. This method of research is referred to as hypothesis testing” (Harland, 2020).

A simple analogy is a chemical experiment conducted in the lab. The first chemical is kept constant (independent variable) and added one drop at a time until a change is noticed in the second chemical (dependent variable). The Chemical Engineer observes the changes and the conditions that caused it (lab temperature, light intensity and so on).

Types of variables

There are three types of variables in an experiment. The *independent variable* is the *one* that is changed or manipulated by the researcher. Just one variable is changed or manipulated so the researcher can track which one caused the observed result. The *dependent variable* is the centre of attention. The researcher pays careful attention to see what changes result in the dependent variable in response to the independent variable. The *controlled variables* are those that must remain constant throughout the process to ensure that the dependent variable is not adversely affected by other factors.



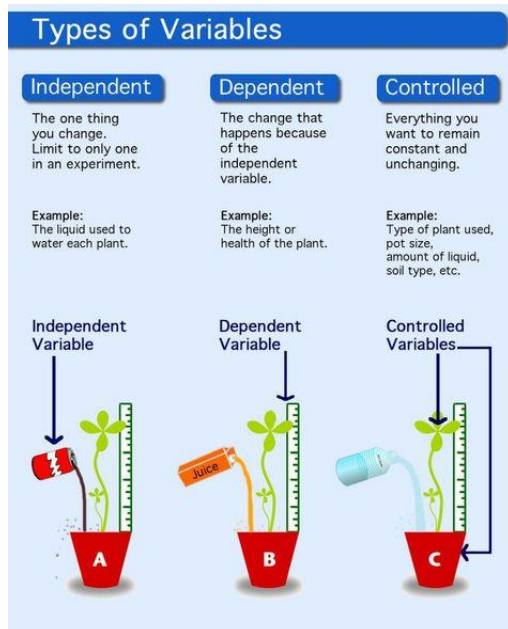


Figure 2: difference between variables in an experiment. Source: Fitch and Tim (2014). Available: <http://grade4ourgwa.weebly.com/4lf-2015-2016/practicting-with-variables>

Some characteristics are:

- The independent variable effects one or more dependent variables.
- A dependent variable changes as a result of changes to the independent variable.
- The experiment manipulates the independent variable (to observe changes in the dependent variable)
- It is good practice to have at most, two independent variables.

The hypothesis

A hypothesis is a proposed explanation for a phenomenon. It is based on the researcher's knowledge, readings and educated assumptions. It can be viewed as a prediction of what might happen.

Example hypotheses:

- H₂O will freeze when the temperature drops below 1 degree Celsius
- Random forest classifier allows faster error prediction in mobile dev projects than neural networks
- Scrum project management results in more user centric software designs

The hypothesis is a statement that is in the form:

Factor A causes B

This hypothesis is formulated such that it can be tested empirically via an experiment. It must also be formulated such that it is *falsifiable* that is, it can be true or false. The aim of hypothesis testing is to try to disprove just as vigorously as approve.

Controlling variables

Experimental research depends on the researcher's ability to isolate factors and show that only *one* factor causes an observed change. This is known as controlling variables.

Approaches for controlling variables include:

- Eliminate outlying factors – for example, the experiment requires experienced Scrum developers, so those not practiced with Scrum will be removed.
- Control groups – two groups are set up with exactly the same characteristics for example, developers with the same expertise and years of experience. For one group, the *control group* there is no manipulation of the of the independent variable. For the second group, the independent variable is manipulated. Effects are measured in each group. Any noted differences in the outcome is a result of the variable that was manipulated.
- Random selection – when there are enough participants chosen on a random basis and those participants that interfere with the outcomes (for example, not enough Scrum experience) cancel each other out across the selection.

Measurement

Measurements need to be performed before and after the experiment. This is referred to as *pre-* and *post-* testing. As an example, should the hypothesis be:

Implementation of anti-X will result in improved team communication

The effect of anti-X can't be effectively measured with just a post-test measurement. Instead, there must have been some pre-test measurement as well to accurately measure change.

Measurement usually includes quantitative data and statistical analysis.

Types of experiments

There are three major types of experiments:

- True experiment: It involves manipulation of the independent variables, pre- and post- test of the dependent variable. "True experiments contain three key features: independent and dependent variables, pretesting and post-testing, and experimental and control groups. In a true experiment, the effect of an intervention is tested by comparing two groups. One group is exposed to the intervention (the experimental group) and the other is not exposed to the intervention (the control group)" (Pressbooks, 2021).
- Pre-experiment: "are called such because they often happen before a true experiment is conducted. Often, researchers want to see if their interventions will have an effect on a small group of people before they seek funding and dedicate time to conduct a true experiment" (Pressbooks, 2021).
- Quasi-experiment: an experimental environment is artificial hence, it might not result in a true reflection. In this type, researchers try to setup the experiment but, in a real-life, natural setting. Quasi means "as if". The control and experimental groups differ not only in terms of the experimental intervention they receive, but also in other unknown or unknowable, ways. Thus, the researcher must try to statistically control for as many of these differences as possible. Because control is lacking in quasi-experiments, there may be several "rival hypotheses" competing with the experimental manipulation as explanations for observed results

5. Case study

The case study focuses on one thing. This could be one SME, one development team, one incident and so on. This single case is studied in-depth to obtain a rich, detailed insight. Popular data collection techniques can include researcher's field notes, interviews, observation, document analysis and questionnaires.

A survey allows for a wide but shallow view of lots of instances of the phenomena. This does not allow much context of the phenomenon. An experiment requires that the phenomenon is divorced from its context in order to ensure the result can only be caused by the researcher's manipulation of the independent variable and not anything else. A case study looks at the case within its real-life context. Oates (2006)

A case study sets out to explore all possible factors and how they interface and influence each other. The researcher asks "how" and "why" certain phenomena occur.

"Case study research is more than simply conducting research on a single individual or situation. This approach has the potential to deal with simple through complex situations. It enables the researcher to answer "how" and "why" type questions, while taking into consideration how a phenomenon is influenced by the context within which it is situated" (Baxter and Jack, 2008).

Characteristics of a case study

The focus is on in-depth and gathering as much details as possible. The case is examined in its natural setting. The researcher tries to understand interfaces, connections and linkages that lead to the complex relationships and processes. There are multiple sources of data and methods for collection to obtain multiple perspectives.

There are some cases whereby the researcher will utilize two or more case studies. The results are compared and the similarities and differences are documented.

Types of case studies

- *Exploratory*. conducted to define the questions or hypothesis for a later, larger study. It helps the researcher to identify a problem or confirm a *hunch*.
- *Descriptive*. is an in-depth investigation and a detailed analysis of a particular case.
- *Explanatory*. one step further than descriptive study. It attempts to explain *why* events happen.

Tutorial

1. Perform a Google scholar search and find 3 IT publications that utilise surveys. Write down the reference and the type of survey that was used.
2. Create an online questionnaire consisting of no more than five questions asking the first year IT student about their learning experiences. You may use any survey tool available online.
3. Given the following study aims, what would be the most appropriate research strategy.
 - a. Developing a random forest model for Long-term follow-up of patients with chronic myeloid leukaemia
 - b. A crash sensor array towards fewer road accident fatalities
 - c. Developing a student centred CX university portal
 - d. Identifying the challenges of Agile adoption in small IT solutions organizations
 - e. Building the first bio-fluid CPU

4. Consider the survey “which is the better platform PyCharm or Pydev” Write five questions for this survey. State with reasons whether each question is quantitative or qualitative?
5. Consider the design science research topic “Developing a model for integration of Green ICT in large retail organizations”. Using Hevner’s methodology, briefly state what needs to be done at each stage of the methodology.
6. Explain the relationship between hypothesis, independent variable and dependent variable.
7. Write down the title and hypothesis for three different research topics.
8. Briefly explain how grounded theory is used in case study research.

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Useful urls

- <https://conjointly.com/blog/data-collection-quantitative-research/>
- <https://conjointly.com/kb/types-of-surveys/>
- <https://explorable.com/types-of-survey>
- <https://www.questionpro.com/blog/types-of-survey/>